

Single-chain Fv Antibodies for Targeting Neurodegenerative Diseases.

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Protein misfolding and aggregation have been considered the common pathological hallmarks for a number of neurodegenerative diseases, including Alzheimer's disease (AD), Parkinson's disease (PD) and Huntington's disease (HD). These abnormal proteins aggregates damage mitochondria and induce oxidative stress, resulting in neuronal cell death. Prolonged neuronal damage activates microglia and astrocytes, development of inflammation reaction and further promotes neurodegeneration. Thus, elimination of abnormal protein aggregates without eliciting any adverse effects are the main treatment strategies. To overcome this, recent studies have deployed single- chain fragment variable antibodies (scFvs) to target the pathological protein aggregates, such as amyloid-beta (A β) peptides, α -synuclein (α -syn) and Huntingtin (Htt). To date scFv has been effective at inhibiting abnormal protein aggregates formation in both in vitro and in vivo model system of AD, PD and HD. Currently active research is still ongoing to improve the scFv gene delivery technology, to further enhance brain penetration, intracellular stability, solubility and efficacy of scFv intrabody.

Characterization of Pairs of Toxic and Nontoxic Misfolded Protein Oligomers Elucidates the Structural Determinants of Oligomer Toxicity in Protein Misfolding Diseases.

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The aberrant misfolding and aggregation of peptides and proteins into amyloid aggregates occurs in over 50 largely incurable protein misfolding diseases. These pathologies include Alzheimer's and Parkinson's diseases, which are global medical emergencies owing to their prevalence in increasingly aging populations worldwide. Although the presence of mature amyloid aggregates is a hallmark of such neurodegenerative diseases, misfolded protein oligomers are increasingly recognized as of central importance in the pathogenesis of many of these maladies. These oligomers are small, diffusible species that can form as intermediates in the amyloid fibril formation process or be released by mature fibrils after they are formed. They have been closely associated with the induction of neuronal dysfunction and cell death. It has proven rather challenging to study these oligomeric species because of their short lifetimes, low concentrations, extensive structural heterogeneity, and challenges associated with producing stable, homogeneous, and reproducible populations. Despite these difficulties, investigators have developed protocols to produce kinetically, chemically, or structurally stabilized homogeneous populations of protein misfolded oligomers from several amyloidogenic peptides and proteins at experimentally amenable concentrations. Furthermore, procedures have been established to produce morphologically similar but structurally distinct oligomers from the same protein sequence that are either toxic or nontoxic to cells. These tools offer unique opportunities to identify and investigate the structural determinants of oligomer toxicity by a close comparative inspection of their structures and the mechanisms of action through which they cause cell dysfunction. This Account reviews multidisciplinary results, including from our own groups, obtained by combining chemistry, physics, biochemistry, cell biology, and animal models for pairs of toxic and nontoxic oligomers. We describe oligomers comprised of the amyloid- β peptide, which underlie Alzheimer's disease, and α -synuclein, which are associated with Parkinson's disease and other related neurodegenerative pathologies, collectively known as synucleinopathies. Furthermore, we also discuss oligomers formed by the 91-residue N-terminal domain of [NiFe]-hydrogenase maturation factor from *E. coli*, which we use as a model non-disease-related protein, and by an amyloid stretch of Sup35 prion protein from yeast. These oligomeric pairs have become highly useful experimental tools for studying the molecular determinants of toxicity

characteristic of protein misfolding diseases. Key properties have been identified that differentiate toxic from nontoxic oligomers in their ability to induce cellular dysfunction. These characteristics include solvent-exposed hydrophobic regions, interactions with membranes, insertion into lipid bilayers, and disruption of plasma membrane integrity. By using these properties, it has been possible to rationalize in model systems the responses to pairs of toxic and nontoxic oligomers. Collectively, these studies provide guidance for the development of efficacious therapeutic strategies to target rationally the cytotoxicity of misfolded protein oligomers in neurodegenerative conditions.

A protein quality control pathway regulated by linear ubiquitination.

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Neurodegenerative diseases are characterized by the accumulation of misfolded proteins in the brain. Insights into protein quality control mechanisms to prevent neuronal dysfunction and cell death are crucial in developing causal therapies. Here, we report that various disease-associated protein aggregates are modified by the linear ubiquitin chain assembly complex (LUBAC). HOIP, the catalytic component of LUBAC, is recruited to misfolded Huntingtin in a p97/VCP-dependent manner, resulting in the assembly of linear polyubiquitin. As a consequence, the interactive surface of misfolded Huntingtin species is shielded from unwanted interactions, for example with the low complexity sequence domain-containing transcription factor Sp1, and proteasomal degradation of misfolded Huntingtin is facilitated. Notably, all three core LUBAC components are transcriptionally regulated by Sp1, linking defective LUBAC expression to Huntington's disease. In support of a protective activity of linear ubiquitination, silencing of OTULIN, a deubiquitinase with unique specificity for linear polyubiquitin, decreases proteotoxicity, whereas silencing of HOIP has the opposite effect. These findings identify linear ubiquitination as a protein quality control mechanism and hence a novel target for disease-modifying strategies in proteinopathies.

The role of melatonin in targeting cell signaling pathways in neurodegeneration.

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Neurodegenerative diseases are typified by neuronal loss associated with progressive dysfunction and clinical presentation. Neurodegenerative diseases are characterized by the intra- and extracellular conglomeration of misfolded proteins that occur because of abnormal protein dynamics and genetic manipulations; these trigger processes of cell death in these disorders. The disrupted signaling mechanisms involved are oxidative stress-mediated mitochondrial and calcium signaling deregulation, alterations in immune and inflammatory signaling, disruption of autophagic integrity, proteostasis dysfunction, and anomalies in the insulin, Notch, and Wnt/ β -catenin signaling pathways. Herein, we accentuate some of the contemporary translational approaches made in characterizing the underlying mechanisms of neurodegeneration. Melatonin-induced cognitive enhancement and inhibition of oxidative signaling substantiates the efficacy of melatonin in combating neurodegenerative processes. Our review considers in detail the possible roles of melatonin in understanding the synergistic pathogenic mechanisms between aggregated proteins and in regulating, modulating, and preventing the altered signaling mechanisms discovered in cellular and animal models along with clinical evaluations pertaining to neurodegeneration. Furthermore, this review showcases the therapeutic potential of melatonin in preventing and treating neurodegenerative diseases with optimum prognosis.

Role of prion protein aggregation in neurotoxicity.

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In several neurodegenerative diseases, such as Parkinson, Alzheimer's, Huntington, and prion diseases, the deposition of aggregated misfolded proteins is believed to be responsible for the neurotoxicity that characterizes these diseases. Prion protein (PrP), the protein responsible of prion diseases, has been deeply studied for the peculiar feature of its misfolded oligomers that are able to propagate within affected brains, inducing the conversion of the natively folded PrP into the pathological conformation. In this review, we summarize the available experimental evidence concerning the relationship between aggregation status of misfolded PrP and neuronal death in the course of prion diseases. In particular, we describe the main findings resulting from the use of different synthetic (mainly PrP106-126) and recombinant PrP-derived peptides, as far as mechanisms of aggregation and amyloid formation, and how these different spatial conformations can affect neuronal death. In particular, most data support the involvement of non-fibrillar oligomers rather than actual amyloid fibers as the determinant of neuronal death.

Toward therapeutic targets for SCA3: Insight into the role of Machado-Joseph disease protein ataxin-3 in misfolded proteins clearance.

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Machado-Joseph disease (MJD, also known as spinocerebellar ataxia type 3, SCA3), an autosomal dominant neurological disorder, is caused by an abnormal expanded polyglutamine (polyQ) repeat in the ataxin-3 protein. The length of the expanded polyQ stretch correlates positively with the severity of the disease and inversely with the age at onset. To date, we cannot fully explain the mechanism underlying neurobiological abnormalities of this disease. Yet, accumulating reports have demonstrated the functions of ataxin-3 protein in the chaperone system, ubiquitin-proteasome system, and aggregation-autophagy, all of which suggest a role of ataxin-3 in the clearance of misfolded proteins. Notably, the SCA3 pathogenic form of ataxin-3 (ataxin-3(exp)) impairs the misfolded protein clearance via mechanisms that are either dependent or independent of its deubiquitinase (DUB) activity, resulting in the accumulation of misfolded proteins and the progressive loss of neurons in SCA3. Some drugs, which have been used as activators/inducers in the chaperone system, ubiquitin-proteasome system, and aggregation-autophagy, have been demonstrated to be efficacious in the relief of neurodegeneration diseases like Huntington's disease (HD), Parkinson's (PD), Alzheimer's (AD) as well as SCA3 in animal models and clinical trials, putting misfolded protein clearance on the list of potential therapeutic targets. Here, we undertake a comprehensive review of the progress in understanding the physiological functions of ataxin-3 in misfolded protein clearance and how the polyQ expansion impairs misfolded protein clearance. We then detail the preclinical studies targeting the elimination of misfolded proteins for SCA3 treatment. We close with future considerations for translating these pre-clinical results into therapies for SCA3 patients.

Engineered Extracellular Vesicles/Exosomes as a New Tool against Neurodegenerative Diseases.

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Neurodegenerative diseases are commonly generated by intracellular accumulation of misfolded/aggregated mutated proteins. These abnormal protein aggregates impair the functions of mitochondria and induce oxidative stress, thereby resulting in neuronal cell death. In turn, neuronal damage induces chronic inflammation and neurodegeneration. Thus, reducing/eliminating these abnormal protein aggregates is a priority for any anti-neurodegenerative therapeutic approach. Although several antibodies against mutated neuronal proteins have been already developed, how to efficiently deliver them inside the target cells remains an unmet issue. Extracellular vesicles/exosomes incorporating intrabodies against the pathogenic products would be a tool for innovative therapeutic approaches. In this review/perspective article, we identify and describe the major molecular targets associated with neurodegenerative diseases, as well as the antibodies already developed against them. Finally, we propose a novel targeting strategy based on the endogenous engineering of extracellular vesicles/exosomes constitutively released by cells of the central nervous system.

Mechanisms of disordered neurodegenerative function: concepts and facts about the different roles of the protein kinase RNA-like endoplasmic reticulum kinase (PERK).

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Neurodegenerative diseases, such as Alzheimer's disease, Huntington's disease, Parkinson's disease, prion disease, and amyotrophic lateral sclerosis, are a dissimilar group of disorders that share a hallmark feature of accumulation of abnormal intraneuronal or extraneuronal misfolded/unfolded protein and are classified as protein misfolding disorders. Cellular and endoplasmic reticulum (ER) stress activates multiple signaling cascades of the unfolded protein response (UPR). Consequently, translational and transcriptional alterations in target gene expression occur in response directed toward restoring the ER capacity of proteostasis and reestablishing the cellular homeostasis. Evidences from in vitro and in vivo disease models indicate that disruption of ER homeostasis causes abnormal protein aggregation that leads to synaptic and neuronal dysfunction. However, the exact mechanism by which it contributes to disease progression and pathophysiological changes remains vague. Downstream signaling pathways of UPR are fully integrated, yet with diverse unexpected outcomes in different disease models. Three well-identified ER stress sensors have been implicated in UPR, namely, inositol requiring enzyme 1, protein kinase RNA-activated-like ER kinase (PERK), and activating transcription factor 6. Although it cannot be denied that each of the involved stress sensor initiates a distinct downstream signaling pathway, it becomes increasingly clear that shared pathways are crucial in determining whether or not the UPR will guide the cells toward adaptive prosurvival or proapoptotic responses. We review a body of work on the mechanism of neurodegenerative diseases based on oxidative stress and cell death pathways with emphasis on the role of PERK.

Molecular pathology in neurodegenerative diseases.

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Neurodegenerative diseases are increasing in prevalence in many countries as the average age of their populations increases, since many of these disorders occur more frequently in elderly individuals, placing an increasing burden on healthcare resources. Most neurodegenerative disorders are associated with

accumulations of abnormal proteins in the central nervous system (CNS), which result in neuronal degeneration and ultimately neuronal death. Recent developments in molecular pathology and genetics have allowed the identification of the abnormal proteins involved in many neurodegenerative disorders and the genes that encode these proteins. This has led to a fuller understanding of the mechanisms of many of these diseases, but this has not so far been accompanied by major improvements in diagnostic tests or treatments for these disorders. Prion diseases are rare neurodegenerative disorders that are associated with the accumulation of a misfolded host protein, the prion protein, in the CNS. Prion diseases have been considered as a paradigm for protein misfolding diseases, but there are significant differences between prion diseases and other neurodegenerative disorders, not least in the transmissible nature of prion diseases. In this review we give an overview of the wide range of neurodegenerative diseases that affect humans, and compare the molecular pathology of prion diseases with other neurodegenerative diseases. The concept of proteinopathy as a common mechanism in neurodegenerative disorders is explored, and we highlight the improvements in diagnosis and management required to improve our treatment of these devastating conditions.

Osmolytes: A Possible Therapeutic Molecule for Ameliorating the Neurodegeneration Caused by Protein Misfolding and Aggregation.

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Most of the neurological disorders in the brain are caused by the abnormal buildup of misfolded or aggregated proteins. Osmolytes are low molecular weight organic molecules usually built up in tissues at a quite high amount during stress or any pathological condition. These molecules help in providing stability to the aggregated proteins and protect these proteins from misfolding. Alzheimer's disease (AD) is the uttermost universal neurological disorder that can be described by the deposition of neurofibrillary tangles, aggregated/misfolded protein produced by the amyloid β -protein ($A\beta$). Osmolytes provide stability to the folded, functional form of a protein and alter the folding balance away from aggregation and/or degradation of the protein. Moreover, they are identified as chemical chaperones. Brain osmolytes enhance the pace of $A\beta$ aggregation, combine with the nearby water molecules more promptly, and avert the aggregation/misfolding of proteins by providing stability to them. Therefore, osmolytes can be employed as therapeutic targets and may assist in potential drug design for many neurodegenerative and other diseases.